Device-leg emulation plan (Catalogizer + Containers submodule)

Field	Value
Revision	1 (draft, research + plan only)
Created	2026-11-12
Proposed repo path	`docs/research/device_emulation_plan_20261112.md` (NOT written to the main checkout)
Status	PROPOSAL, needs independent review (§11.4.142 / §11.4.209) and operator sight
Operator decision cited	2026-11-12 (relayed): no TV, phone or second machine connected; fan out emulators of all kinds for device legs via the Containers submodule and extend it upstream (§11.4.74, §11.4.76, §11.4.161)
Author	research subagent (Opus); no code, no image pull, no emulator started

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All `file:line` references are into `constitution/submodules/containers` at commit `5ea870a` (equal to `origin/main`: `git ls-remote origin main` = `5ea870a5837097077abcf04caab29550dca2fa5f`, measured 2026-11-12). The flat copy `submodules/containers` is at `22b4970`, an ancestor 9 commits behind (`git merge-base --is-ancestor` true, `git log 22b4970..5ea870a | wc -l` = 9). Upstream work starts from `5ea870a`.

1. Measured host facts (2026-11-12 ~23:48 CET, read-only probes)

Fact	Value	Command
CPU	AMD Ryzen 7 2700X, 16 threads, `svm` present (16 hits)	`/proc/cpuinfo`, `nproc`
KVM	`/dev/kvm` `crw-rw----+ root kvm`, ACL `user:milosvasic:rw-`; readable and writable by this user; module `kvm_amd`, nested = 1	`ls -l`, `getfacl`, `test -r/-w`, `/sys/module/kvm_amd/parameters/nested`
RAM	30 GiB total, ~17 GiB available; memory PSI some avg10 = 1.08	`free -g`, `/proc/pressure/memory`
Load	26.4 / 26.7 / 25.4 on 16 threads	`uptime`
Disk	`/` 1.8T, 97% used, 58 GiB free	`df -h`
Process limit	`ulimit -u` = 123694	
Rootless podman	5.7.0, rootless = true, backend netavark, pasta present; 45 containers running; rootless bridge networks exist (`podman network ls`)	`podman info`, `podman ps -q | wc -l`
`/dev/net/tun`	present	
Binder (redroid)	`/dev/binder*` and binderfs ABSENT; `binder` not in `/proc/filesystems`	`ls`, `grep`

Fact	Value	Command
Host Android SDK	~/Android/Sdk with emulator 37.1.11, system-images;android-34;android-tv;x86 (8.2 GiB, TagId android-tv, rev 3) and system-images;android-34;default;x86_64 (1.7 GiB); ~/.android/avd empty	ls, source.properties, du -sh
Host tools	adb (SDK), scrcpy (/usr/bin/scrcpy), smbd (/sbin/smbd), rclone present; qemu-system-x86_64, qemu-img, swtpm, Xvfb, xvfb-run, weston ABSENT	command -v
Local images	53 localhost/* images; ZERO matching tauri|android (control: same pipeline lists localhost/catqa3-*); dxflrs/garage:v1.0.1, atmoz/sftp, alpine:3.20, debian:bookworm-slim, debian:trixie-slim present	podman images

Consequences: KVM works rootless for this user (named-user ACL, which is exactly the case pkg/emulator/containerized.go:78-95 handles with --userns=keep-id). CPU is the scarce resource (load 26 on 16 threads); RAM allows at most two Android AVDs at a time inside the §12.6 60% ceiling (18 GiB) with the other sessions' footprint. Disk (58 GiB) rules out Windows images and new multi-GB emulator images until the operator prunes. The Tauri QA image catalogizer-tauri:qa is no longer on the host and must be rebuilt before any Tauri live run.

2. Inventory: what the Containers submodule already does

Capability	Evidence	Notes
Android emulator orchestration, two runners (host-direct, containerized)	pkg/emulator/android.go:151 (AndroidEmulator), pkg/emulator/containerized.go:286 (Containerized), :427 Boot, :540 WaitForBoot, :791 Install, :815 RunInstrumentation, :883 Teardown	Emulator interface pkg/emulator/types.go:345
Runner policy per OS	pkg/emulator/accel.go:126 AccelProfileForOS, :184 ResolveRunner, :219 GateEligibleForOS (gate-eligible only when runner == OS profile runner)	On Linux the gate-eligible runner is containerized; the host-direct run Catalogizer did in docs/qa/android-boot-20261105 is NOT gate-eligible by this rule
Form factor field incl. tv	pkg/emulator/types.go:31-44 (AVD.FormFactor "phone", "tablet", "tv", ...)	Metadata only: no TV-specific code path (grep "tv"|leanback|tv_1080p in non-test emulator code: 0 hits apart from the comment)
System-image cache incl. android-tv URL parsing	pkg/emulator/android.go:274 ensureSystemImageViaCache, :344 parseSystemImageURL (example sys-img/android-tv/x86-30_r4.zip, :337)	
KVM passthrough + rootless KVM fix	pkg/emulator/containerized.go:37 kvmAvailable (os.Stat only), :93-102 --userns=keep-id + --device /dev/kvm	Stat does not prove the device is openable; pkg/vm/qemu.go:163-164 opens /dev/kvm O_RDWR instead (the stronger check)
Argv injection guard	pkg/emulator/containerized.go:111-126 (-- before image)	
ADB hygiene, phantom TCP, boot diagnostics	pkg/emulator/adb_hygiene.go:60 ParseADBDevicesLine, :134 ResetADBHygiene, :223 CaptureBootDiagnostic	
Zombie reaping by real cmdline	pkg/emulator/cleanup.go:75 Cleanup, :234 KillByPort, pkg/emulator/cleanup_os.go:42-64 (/proc walker)	Matches §11.4.196(D) discipline
Network shaping (edge..ethernet)	pkg/emulator/network.go:34, :93 applyNetworkConditions; pkg/vm/network.go (duplicate table)	Emulator-console shaping only, not a second network host
Screenshot	pkg/emulator/screenshot.go:24 CaptureScreenshot; pkg/vm/screenshot.go	No screen recording: screenrecord 0 files, scrcpy 0 files (control screencap = 3 files)
Release cold-start canary	pkg/emulator/canary.go:139 RunCanary (activity RESUMED + logcat FATAL-free), cmd/emulator-canary	Launch via am start; no leanback launcher intent
Matrix runner + attestation	pkg/emulator/matrix.go:525 RunMatrix, :622 writeAttestation, cmd/emulator-matrix	
Baked emulator image	pkg/emulator/Containerfile (ARGs API_LEVEL, ABI, TARGET_TYPE, lines 36-39); images/android-test/Dockerfile (google_apis x86_64 API 28-36) + images/android-test/avds/config_api*.ini (tag.id=google_apis)	No TV AVD config shipped; image base ghcr.io/vasic-digital/android-emulator-base:latest (mutable tag) not present locally

Capability	Evidence	Notes
QEMU Linux VMs (alpine/debian/fedora, x86_64/aarch64/riscv64)	`pkg/vm/qemu.go:189` Boot (`-m 2048 -smp 2`, user-net with SSH hostfwd `:201`, `-snapshot` on cold boot `:221`), `pkg/vm/types.go:25` `VMTarget`, `pkg/vm/clients.go` (real SSH/QMP), `cmd/vm-matrix`	Launches `qemu-system-*` ON THE HOST (`pkg/vm/qemu.go:173`); the host has no QEMU, so this path cannot run here today
Image cache with sha256 manifest	`pkg/cache/manifest.go:28` `ImageEntry` (sha256 + size REQUIRED)	
Cuttlefish	`pkg/cuttlefish/types.go:132-176`: requires ROOTFUL + `--privileged` + `--network host`	Excluded by the operator constraints (rootless, no privileged)
macOS/iOS	`pkg/vm/macos/macos.go:1-29` (Tart, Apple-Silicon macOS host only), `pkg/vm/ios`, `pkg/applesim`	Not usable on this Linux host
Genymotion	`pkg/genymotion/genymotion.go`	Commercial/cloud; not considered
Windows	`pkg/crossbuild/windows_wine.go` + `windows_wine.Containerfile` (Wine + JDK + Gradle)	BUILD backend only; no Windows runtime/VM target (grep `windows` in `pkg/vm/*.go` non-test: 0 hits)
Generic run primitive	`pkg/runtime/run.go:77-88` `runOptions` (Name, Entrypoint, WorkDir, User, Network, Env, Volumes, ExtraArgs), With* options `:119-187`	No typed memory / cpus / pids / device / userns / publish / label / digest options; this is Catalogizer's recorded deviation (b) in `scripts/containers/run_pinned.sh:44-45`
Remote distribution, SSH tunnels, overlay	`pkg/network/overlay.go:14-173` (`TunnelOverlay` over SSH), `pkg/network/tunnel.go`, `pkg/volume/nfs.go`, `sshfs.go`, `rsync.go`, `pkg/remote`	Overlay is SSH-tunnel based (needs a real remote host); no rootless virtual-LAN-peer primitive
Protocol-server fixtures (SMB/FTP/WebDAV/NFS/S3)	none: `grep -rli 'samba|smbd|webdav|vsftpd|ftpd|minio|garage'` over Go/Dockerfile/yml found only `pkg/policy/policy.go` (an unrelated mention)	Catalogizer keeps its own `docs/qa/test-servers/up.sh` (`cattsrv-*`)
Desktop GUI runtime (Xvfb/VNC/WebKit)	none in `pkg/cmd/images` (only `xvfb` as a build dep in `pkg/crossbuild/windows_wine.Containerfile:41`)	Catalogizer keeps its own `scripts/tauri/run-qa.sh` (Xvfb + openbox + x11vnc + noVNC, `--memory=2G --cpus=2 --pids-limit=1024`)
Resource limits on emulator containers	none: `grep -rni 'memory|cpus'` over `pkg/emulator`, `pkg/cuttlefish`, `pkg/vm` non-test = 0 hits (control `kvm` = 33 hits in `containerized.go`)	Host-safety gap (§12.6)

3. Catalogizer device legs: what emulation can and cannot prove

Sources: `docs/CONTINUATION.md` Rev 20 (lines 16, 26-27, 37, 67, 87-91, 98-100) and `docs/FOLLOW_UPS_NEXT_PHASE.md` (lines 1503-1520 Android/Tauri, 1425/1578/1614 second LAN host, 605/1245/1384 Synology, 1617-1699 Tauri wizard).

Rule applied. Under §11.4.289 an emulator running the REAL artifact (the real APK, the real Tauri binary) against the REAL API is LIVE-class evidence for claims scoped to the emulated target; it is not a test double. Two limits bind every row below:

- The record must carry the §11.4.289(D) fields: artifact sha256, AVD/system-image identity (`ro.build.fingerprint`, system-image `source.properties`), emulator version, container image digest, endpoint actually reached.
- §11.4.298(B), verbatim: "An emulator, a simulator, a host-only run, a replayed or copied record, or a run on hardware of another product is never this run." So no emulated run makes a product `RELEASABLE` under the release gate; Android, Android TV and any hardware-bound product stays `BLOCKED-NO-HARDWARE` for deploy/tag purposes until real hardware is connected. Emulation raises PROVISIONAL items to VERIFIED only for the emulated-target claim, and the item text must say which target.

Leg (item ids)	Emulated target	Claim an emulated run CAN support	Claim it CANNOT support (stays SKIP `hardware_not_present` / open)	Verdict
Tauri desktop, Linux (CAT-TAURI-NATIVE-1, TAURI-FU-7)	real Linux binary in WebKitGTK under Xvfb in a rootless container	first-run URL dialog, confirm-on-URL-change, login, SSRF refusal, close-X, all driven see-before-type (§11.4.193) with OCR/screenshots, N>=3	GPU compositing (WEBKIT compositing disabled), Wayland, HiDPI, real desktop environments, tray/notifications on real DEs	COVERABLE (needs image rebuild)

Leg (item ids)	Emulated target	Claim an emulated run CAN support	Claim it CANNOT support (stays SKIP `hardware_not_present` / open)	Verdict
Tauri installer wizard (CAT-TAURI-INSTALLER-1, TAURI-R1-FU-7)	same	GREEN run of the Save Configuration sequence on the rebuilt wizard, same target as the RED	install onto a real user session, OS integration (menu entries, file associations)	COVERABLE
Tauri Windows build	QEMU Windows guest (not implemented)	only with an operator-supplied, licensed Windows image recorded as DATA; WebView2 runtime behaviour	anything without that image; Wine does NOT count (WebView2 under Wine UNCONFIRMED, and Wine is not Windows)	SKIP until operator provides image + disk; reason `topology_unsupported`
Tauri macOS build	none on this host	none	everything: macOS on non-Apple hardware is not permitted by Apple's licence (operator to confirm the reading); `pkg/vm/macos` needs an Apple-Silicon macOS host	SKIP (structurally impossible on this host)
Android phone (CAT-ANDROID-LOGIN-1, AND-FU-3/4/6)	AVD `pixel_6`, API 34 `default;x86_64` (image on host), KVM	install, launch, server-URL entry, real qa_admin login via secret-safe input, browse/search/play against the real API, AND-FU-6 "Connected" on unreachable server reproduction	ARM-only native code paths, real touch latency, radios/Wi-Fi roaming, battery, OEM skins, HW codecs, real performance figures, release gate	COVERABLE for the emulated-target claim
Android TV (AND-FU-2, TV player items)	AVD `tv_1080p`, API 34 `android-tv;x86` (image on host, 8.2 GiB)	leanback launch, D-pad navigation/focus, login, browse, playback with software decode	HDMI/CEC, HW decode (HEVC/HDR/Dolby), real remote (IR/BT), TV OEM firmware, 4K output timing	COVERABLE for the emulated-target claim
Display access from a second machine (AND-FU-4)	n/a	none: needs a second machine by definition	the SSH-tunnel + scrcpy path from another host	SKIP `hardware_not_present`
"Anton" SMB second machine (CONTINUATION 87-91: "Anton.local IS this host") and bare-IP SMB	rootless netavark bridge with a peer container (own netns, own IP) running a real Samba; optional containerized QEMU Linux guest for a second kernel	client works against an SMB server at a different IP over a virtual L2 link (not loopback, not the API's own netns), bare-IP addressing, port 445 vs non-445, cut-recovery by stopping the peer, N>=3	a physical NIC, switch, router, MTU/Wi-Fi loss, a Windows SMB server, Synology's Samba build	COVERABLE as "virtual LAN peer"; physical second-host claim stays SKIP
NFS / FTP / FTPS / SFTP / WebDAV / S3 on a second host	same peer pattern: userspace NFS (rclone/ganesha), vsftpd or pure-ftpd, atmoz/sftp, a WebDAV server, Garage (image on host)	protocol client behaviour at a non-local IP	kernel-NFS server behaviour (needs root), client-side NFS kernel mount (needs root; the earlier probe used sudo)	COVERABLE for userspace servers; kernel-NFS SKIP
TRUSTED_PROXIES / rate-limit from a second LAN host (STK3-FU-1, QA-MON-FU-4, TP-FU-3)	peer container on a separate bridge sending through the nginx front door	API sees a distinct client IP arriving over a veth/bridge path, XFF handling with a non-primary source	"real remote NIC" (stays UNCONFIRMED, as the 2026-11-06 substitute run already states)	PARTIAL, label precisely
Synology SMB/FTP/NFS/WebDAV, idle-deadline, cut-recovery (CAT-SYN-LIVE-1, ST-NFS, LV3-2)	none	none: generic Samba/FTP fixtures prove nothing about the Synology firmware; DSM emulation (XPEnology-class) is not pursued (DSM licence is tied to Synology hardware; operator to confirm)	all Synology claims	SKIP `hardware_not_present` (drives OFF since 2026-11-05)
Real phone / real TV hardware	none	none	every hardware claim and the §11.4.298 release gate	SKIP `hardware_not_present`

Already proven once (do not re-claim): both debug APKs built in a rootless JDK 17 container and the phone and TV AVDs booted headless with KVM on the host (`docs/qa/android-boot-20261105/README.md`), via the host-direct runner and project-local commands, not via the Containers submodule.

4. Gap list for the Containers submodule

Each gap: proposed API, tests written first (RED captured on the current code, §11.4.224 / §11.4.292), paired mutants, size (non-test LOC + test LOC, estimate), host-safety notes. All new code is project-agnostic (§11.4.28): Catalogizer names, ports and IPs are consumer DATA.

G1. Typed resource and device options in the run primitive (pkg/runtime)

- API: `WithRunMemory(bytes)`, `WithRunMemorySwap(bytes)` (default = memory, i.e. no swap), `WithRunCPUs(float)`, `WithRunPidsLimit(n)`, `WithRunDevice(path)` (validated: must exist, must be in an allowlist the caller passes; `/dev/kvm` opened O_RDWR first), `WithRunUserns("keep-id")` (podman only; docker returns a typed error), `WithRunPublish(hostIP, hostPort, ctrPort, proto)` (hostIP REQUIRED, `0.0.0.0` refused unless `WithRunPublishAllInterfaces()` is explicit), `WithRunLabel(k,v)`, `WithRunImageDigest(ref)` (refuses a mutable tag when digest pinning is requested). Never `--privileged`, never `--network host`: both refused at the builder with a typed error.
- Tests first: arg-builder table tests (each option present/absent), refusal tests (privileged, network host, 0.0.0.0 publish, docker + userns), a podman-real test gated on podman presence that runs `alpine:3.20` (already on host) with `--memory` and reads `/sys/fs/cgroup/memory.max` inside to prove the limit is real.
- Mutants (>=4): drop the memory flag; drop the `--` before image; allow `--privileged`; treat a tag as digest.
- Size: ~250 + ~400 LOC. Unblocks Catalogizer deviation (b) (`scripts/containers/run_pinned.sh:44`).

G2. Host-safety admission + emulator envelope (pkg/emulator, shared helper in pkg/policy or new pkg/hostbudget)

- API: `hostbudget.Admit(req Need) (Verdict, Evidence)` with `Need{MemBytes, CPUs, Pids, NeedsKVM}`; verdict closed set `fits | does_not_fit | undetermined` (mirrors §11.4.291 vocabulary); inputs `/proc/meminfo` MemAvailable/MemTotal, `/proc/pressure/memory` and `/proc/pressure/cpu`, loadavg vs nproc, `ulimit -u`, `/dev/kvm` O_RDWR probe (read-only intent: open+close, no ioctl). Reserve and thresholds are consumer DATA; defaults = §12.6 60% ceiling, `MemAvailable - need >= max(4 GiB, 15% MemTotal)`. `ContainerizedConfig` gains `Memory`, `CPUs`, `PidsLimit` (host-adaptive defaults; emulator default memory = AVD RAM + 1.5 GiB overhead, CPUs = min(3, 0.25*nproc)); `Boot` refuses on `does_not_fit` or `undetermined` with a typed error and writes the evidence JSON. Replace `kvmAvailable` stat (`pkg/emulator/containerized.go:37`) with the O_RDWR probe from `pkg/vm/qemu.go:163`.
- Tests first: fake `/proc` fixtures (golden-good fits, golden-bad does_not_fit, unreadable PSI -> undetermined, carrier: high load but low memory need is still judged by both axes); KVM probe with a temp char-device stand-in is not possible, so test the error-mapping seam plus one real probe on hosts with `/dev/kvm`.
- Mutants: stat instead of open; ignore PSI; treat undetermined as fits; drop `--memory` from run args.
- Size: ~300 + ~500 LOC.

G3. Host-SDK-mount containerized emulator (no multi-GB image pull)

- Problem: the gate-eligible Linux runner needs an emulator image; the base `android-emulator-base:latest` is absent and disk is at 97%. The host already has the SDK, emulator 37.1.11 and both API 34 system images.
- API: `ContainerizedConfig.SDKMount{HostPath, ReadOnly: true}` and `AVDHome` (scratch dir, rw): run a slim image (Debian bookworm-slim, already on host, plus the emulator's runtime libs: libpulse0, libnss3, libxcomposite1, libxcursor1, libxi6, libxtst6, libasound2, socat) with the host SDK bind-mounted read-only at `/opt/android-sdk`, AVDs created by `avdmanager` inside the container into the scratch AVD home. New `images/android-emulator-slim/Containerfile`; image built locally (one ~150 MB layer, no SDK inside). Record the system-image `source.properties` and `ro.build.fingerprint` in `BootResult`.
- Tests first: arg-builder tests (mount ro, AVD home rw, no write mount on the SDK), avdmanager command builder for `-d pixel_6 -k "system-images;android-34;default;x86_64"` and `-d tv_1080p -k "system-images;android-34;android-tv;x86"`; live boot test gated on KVM + host SDK + admission `fits`.
- Mutants: rw SDK mount; wrong `-k` tag for TV; skip fingerprint capture; omit `--userns=keep-id`.
- Size: ~350 + ~450 LOC + Containerfile ~40 lines.

G4. Android TV first-class (pkg/emulator)

- API: `FormFactorTV` constant; `TVProfile{Device: "tv_1080p", Tag: "android-tv", ABI: "x86"}`; canary option `LaunchCategory: "android.intent.category.LEANBACK_LAUNCHER"`; D-pad helpers `DPad(ctx, keys...)` mapping to `input keyevent KEYCODE_DPAD_UP/DOWN/LEFT/RIGHT/CENTER`, `KEYCODE_BACK`, `KEYCODE_HOME`; `FocusedNode()` via `uiautomator dump` and a parser that returns the focused node (see-before-type, §11.4.193). `images/android-test/avds/config_api34_tv.ini`.
- Tests first: keyevent argv tables, uiautomator XML fixtures (golden focused node, no focused node -> error, carrier: `focusable=true` but not `focused`), canary intent builder for leanback.
- Mutants: phone launcher category for TV; DPAD_CENTER mapped to ENTER; parser accepting `focusable`.
- Size: ~300 + ~450 LOC.

G5. Recording and frame evidence (pkg/emulator, pkg/desktopgui)

- API: `StartRecording(ctx, serial, maxDur) (Recorder, error)` using `adb shell screenrecord --time-limit` (180 s cap per segment, rotate) with pull + sha256; optional scrcpy backend when present on host (`scrcpy --no-window --record`); `CaptureFrame` returns PNG + sha256 + timestamp + build fingerprint.
- Tests first: argv builders, segment rotation, sha256 file writing, failure on 0-byte file (FALSE-NULL guard).
- Mutants: accept empty file; skip rotation; drop time limit; wrong serial flag.
- Size: ~250 + ~350 LOC.

G6. Secret-safe text input (pkg/emulator)

- Problem: AND-FU-3, `adb shell input text <password>` exposes the secret in the HOST process table.
- API: `TypeSecret(ctx, serial, secretFile string)`: pushes the 0600 file to `/data/local/tmp/<random>` via `adb push` (path in argv, value not), runs `adb shell 'input text "$(cat F)"; rm -f F'` with the command string free of the value, then verifies with `uiautomator dump` that the field is non-empty and masked. Secret never logged; evidence redacts.
- Tests first: argv never contains the secret (property test over random secrets), file removed on success and on failure path, refusal when the file is not 0600.
- Honest limit: the value is briefly in the device's own shell process; acceptable for an emulator (UNCONFIRMED whether the operator accepts it).
- Mutants: secret in argv; no rm; mode check skipped; no post-type verification.
- Size: ~150 + ~250 LOC.

G7. Desktop GUI runtime for Linux apps (new pkg/desktopgui)

- API: `desktopgui.Session{Image, AppPath(ro), Screen, BindIP(default 127.0.0.1), VNCPasswordFile(0600), Memory, CPUs, PidsLimit}` with `Start`, `Screenshot` (xwd -> PNG + sha256), `OCR` hook (consumer-supplied), `Type`/`Click` via xdotool (see-before-type: screenshot before and after), `WebDriver()` endpoint when `tauri-driver` + `WebKitWebDriver` are in the image, `Stop`. New `images/desktopgui-webkit/Containerfile` (Debian bookworm-slim + Xvfb + openbox + x11vnc + novnc + websockify + xdotool + libwebkit2gtk-4.1). Port of the logic in Catalogizer `scripts/tauri/run-qa.sh` (Catalogizer then calls this package; its script becomes a thin wrapper).
- Tests first: arg builder (ro app mount, loopback bind default, LAN bind needs explicit flag, no host network), password-file mode check, screenshot non-blank check (golden blank frame must FAIL).
- Mutants: bind 0.0.0.0 by default; accept blank screenshot; rw app mount; missing `--pids-limit`.
- Size: ~500 + ~600 LOC + Containerfile ~40 lines.

G8. Virtual LAN peer + protocol-server fixtures (extend pkg/network, new pkg/fixtures)

- API: `network.CreateBridge(name, subnet)` (rootless netavark, internal by default), `fixtures.Peer{Network, IP, Services []Service}` with typed services `SMB{Share, Path, Users from 0600 file}`, `NFSUser{Export, RO}` (userspace), `FTP{Explicit/Implicit TLS}`, `SFTP`, `WebDAV`, `S3Garage`; each with a protocol-level health check (SMB: `smbclient -L` or a Go SMB negotiate; NFS: MOUNT/NULL RPC; FTP: banner + login; WebDAV: PROPFIND 207; S3: signed ListBuckets) and a fault API `Cut()` / `Restore()` (disconnect the peer from the bridge) for cut-recovery tests. Corpus mounted read-only except a declared `rw/` subdir. Ports: inside the bridge the real ports (445, 2049, 21) are used; no host publish unless asked.
- Tests first: arg builders, health checks against fixtures that answer wrongly (golden-bad: FTP banner but login refused must not be "healthy"), cut/restore state machine, real-podman test on alpine.
- Mutants: health = TCP connect only; publish on 0.0.0.0; corpus rw; Cut() that does not detach.
- Size: ~800 + ~1000 LOC + 4 small Containerfiles (Samba, vsftpd, WebDAV, NFS-userspace; reuse `atmoz/sftp` and `dxflrs/garage:v1.0.1` already on host).
- Honest limit: proves "different IP over a virtual L2 link", never a physical second machine.

G9. Containerized QEMU runner (pkg/vm)

- Problem: `pkg/vm/qemu.go:173` launches `qemu-system-*` on the host; the host has none.
- API: `vm.NewContainerizedQEMU(cfg{Image, Memory, CPUs, KVM})` implementing the existing `VM` interface (`pkg/vm/types.go:179`); `--device /dev/kvm`, `--userns=keep-id`, qcow2 base read-only + `-snapshot`, networking either user-net with SSH hostfwd (existing) or `-netdev tap` inside the container attached to a G8 bridge (container has `/dev/net/tun` via `--device`, no privilege). `-m`/`-smp` from config instead of the hardcoded `2048`/`2` (`pkg/vm/qemu.go:196-197`).
- Tests first: arg builder parity with host runner, refusal without admission, live boot of `alpine-3.20-x86_64` gated on KVM (image fetched only through `pkg/cache` with sha256; not now: disk).
- Mutants: hardcoded memory; missing `-snapshot`; rw base image; tap without tun device.
- Size: ~400 + ~500 LOC + Containerfile.

G10. Windows guest target (pkg/vm, later)

- API: `VMTarget{Distro: "windows", ...}` + OVMF + swtpm (TPM 2.0) inside the G9 container; image ONLY from an operator-supplied path with licence DATA (`licence: evaluation|retail`, expiry); never auto-download; boot readiness via WinRM or OpenSSH-for-Windows.
- Blocked now: disk (Windows image 20-30 GiB vs 58 GiB free), licence decision (operator), host load.
- Size: ~600 + ~600 LOC. Record SKIP reason `topology_unsupported` until the operator supplies an image.

G11. Reachable-endpoint resolver (pkg/emulator, pkg/desktopgui)

- Problem: inside a rootless pasta container the host LAN IP resolves to the container (`scripts/tauri/run-qa.sh:10-12`); inside the Android emulator the host is `10.0.2.2`. Every run must record which URL it actually used.
- API: `ReachableAPI(target, hostPort) (url, evidence)` returning `http://10.0.2.2:<p>` (emulator), `http://host.containers.internal:<p>` (container), or a G8 bridge IP, with a probe from inside the target (curl/`adb shell toybox nc`) recorded as §11.4.289(D) endpoint identity.
- Size: ~150 + ~250 LOC.

G12. Device leases for emulated devices (reuse §11.4.119 / §11.4.176 pattern)

- API: lease keyed by stable AVD name / container name, all-or-nothing, TTL-reaped by `kill -0` + cmdline, so parallel agents never drive one emulator. Size ~200 + ~300 LOC. May reuse an existing lease engine in the constitution (UNCONFIRMED whether one is importable by reference; check before writing).

Host-safety limits for every gap (binding): rootless only; `--privileged`, `--network host`, Cuttlefish and redroid (needs binder kernel module, absent) excluded; `/dev/kvm` touched only by open+close probe and by `--device` passthrough; every container carries `--memory`, `--memory-swap` = memory, `--cpus`, `--pids-limit`; whole runs wrapped in `systemd-run --user --scope -p MemoryMax=… -p MemorySwapMax=0`; at most 2 AVDs concurrently on this host (2 x ~4.5 GiB); no run while load > ~12 (CONTINUATION dispatch rule); no image pull until the operator approves disk use; no nested privileged containers; no emulator bound to 0.0.0.0 (adb/console unauthenticated).

5. Ordered work plan

Each step: one Sonnet author in a scratch clone of `vasic-digital/Containers` at `5ea870a` (fast-forward only, §11.4.288 vehicle rules), test-first, >=4 own mutants, then Opus `xhigh` review (§11.4.209), budget 5 rounds (§11.4.276(G)). Steps on one line run in parallel (disjoint files).

Order	Steps	Depends on	Can run under current load/disk?
1	G1 (pkg/runtime) ‖ G2 (hostbudget + emulator envelope) ‖ G11 (resolver)	none	YES: unit + alpine-only real tests
2	G3 (SDK-mount slim emulator) ‖ G4 (TV) ‖ G6 (secret input)	G1, G2	code YES; live boot only when load < ~12
3	G5 (recording) ‖ G7 (desktopgui)	G1, G2	code YES; image build needs base already on host (bookworm-slim) plus apt layer (network, ~300 MB)
4	G8 (virtual LAN peer + fixtures)	G1	code YES; Samba/vsftpd images need apt layers
5	G9 (containerized QEMU) ‖ G12 (leases)	G1, G2	code YES; live VM needs a qcow2 via pkg/cache
6	Catalogizer adoption: bump `submodules/containers` + constitution pin to the new tag; `scripts/tauri/run-qa.sh`, `docs/qa/android-boot-*` commands and `docs/qa/test-servers/up.sh` become thin wrappers; remove deviation (b)	1-5 landed upstream	YES
7	Live legs, N>=3 each: Android phone login (AND-FU-3, AND-FU-6 repro), Android TV D-pad + playback, Tauri desktop GREEN, wizard GREEN (TAURI-R1-FU-7), SMB/NFS/FTP/WebDAV/S3 at a peer IP, bare-IP SMB, TRUSTED_PROXIES from peer	6	only when load < ~12 and disk freed
8	G10 (Windows)	operator: image + licence + disk	NO (blocked)

Size total (estimate, non-test + test): ~4.4k + ~5.6k LOC plus 7 small Containerfiles; steps 1-5 ~ 12-16 author dispatches including review rounds.

Operator decisions needed before step 2 live runs: (a) disk: approve pruning or a target for the ~300 MB- 1 GB of new image layers; (b) whether the in-device secret exposure of G6 is acceptable; (c) Windows image + licence (G10) or permanent SKIP; (d) confirm the macOS and Synology-DSM licence readings above.

6. Findings outside scope and UNCONFIRMED list

Findings (not fixed here, for registration as follow-ups):

- F1. `helix-deps.yaml:93` records `ref: bfd0e98` for `containers`, but the gitlink is `22b4970` (`git ls-tree HEAD submodules/containers`); `bfd0e98` is an ancestor (2 commits). Manifest drift (§11.4.31 / §11.4.233(G)).
- F2. Two checkouts of the same repo at different commits: `submodules/containers` `22b4970` and `constitution/submodules/containers` `5ea870a`. Decide which one Catalogizer code builds against (§11.4.28(C) one source of truth).
- F3. `pkg/emulator/containerized.go:37` judges KVM by `os.Stat`; `pkg/vm/qemu.go:163` already uses the stronger O_RDWR open. Inconsistent guard (§11.4.201). Covered by G2.
- F4. No resource limits on emulator/QEMU containers (control-needled 0 hits). Covered by G1/G2.
- F5. Catalogizer's Android boot and Tauri QA runs use project-local commands, not the Containers submodule (§11.4.74 / §11.4.76 deviation). Covered by step 6.
- F6. `images/android-test/Dockerfile` builds FROM a mutable tag `android-emulator-base:latest` (§11.4.264 digest pinning).

UNCONFIRMED:

- Exact host RAM footprint of a running TV AVD under container + Xvfb-free headless mode on this host (earlier run used `MemoryMax=5G` per emulator; not re-measured).
- Whether the host emulator 37.1.11 binary runs inside Debian bookworm-slim with only the listed libs (G3 assumption; needs the first RED/GREEN).
- Whether rootless netavark allows `--device /dev/net/tun` + tap inside a container for QEMU (G9).
- WebView2 behaviour under Wine; legal reading of macOS and Synology DSM licences (operator).
- Whether a reusable lease engine exists in the constitution for G12.
- Whether upstream `vasic-digital/Containers` has unpushed work on other mirrors (only `origin` checked).